

Signal-Informed Temporal Routing for Vinyl Defect Regime Detection

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Abstract—Vinyl restoration systems must distinguish isolated clicks, short bursts, dense crackle, and overlapping damage before selecting a repair operation. We present a lightweight two-stage detector in which signal-informed sparse, burst, and dense experts produce complementary defect evidence, and a temporal backend converts that evidence into stable repair regimes. The backend factors the five-way decision hierarchically, applies a validation-only mixed regime gate, and decodes with validation-selected transition penalties that outperform a maximum-likelihood transition matrix on the same emissions. On a source-separated synthetic benchmark of 597 non-overlapping 15 s excerpts drawn from 21 recordings, the held-out system reaches 0.730 pooled five-class macro F1 (95% CI 0.694–0.761), an improvement of 0.043 over a flat frame-level router (paired bootstrap $p = 0.001$). Clean and dense frames are highly reliable, while sparse, burst, and mixed frames remain far more ambiguous.

Index Terms—audio restoration, vinyl defects, expert ensembles, temporal routing, sequence decoding, archival audio

I. INTRODUCTION

Archival audio restoration is difficult because degradation is not a single noise process. A digitized record can contain isolated clicks, wide pops, short scratches, dense crackle, low-frequency disturbances, and overlapping damage. These artifacts differ in duration, density, spectrum, and repair assumption. A narrow click may be repaired by local interpolation, whereas a scratch requires region inpainting and dense crackle a different denoising strategy. A restoration system therefore needs to detect damage and, at the same time, route each time region to the appropriate repair mode.

Classical declipping assumes a localized impulsive corruption model,

$$x[n] = s[n] + c[n], \quad (1)$$

where $s[n]$ is clean audio and $c[n]$ a sparse impulsive component [1]. This motivates detect-and-replace pipelines but becomes unreliable when micro impulses occur densely, defects overlap, or musical transients resemble damage. A single declipping score cannot decide whether a region should be interpolated, inpainted, denoised, or left unchanged.

This paper formulates vinyl defect analysis as a repair-oriented regime routing problem. The front end produces

three evidence streams, namely candidate-level sparse evidence for isolated clicks and pops, frame-level burst evidence for wide pops and scratches, and frame-level dense evidence for crackle. A learned hierarchical backend then combines these streams with confidence, interaction, and local temporal features, and a validation-selected Viterbi decoder produces a stable five-class sequence over clean, sparse, burst, dense, and mixed regimes.

Our contributions are fourfold. First, we design a two-stage defect-regime detector whose front end pairs signal-informed candidate and frame features with three task-specific experts. Second, we show that factoring the decision hierarchically and decoding it with validation-selected transition penalties improves macro F1 over both a flat router and a maximum-likelihood transition matrix, under both the five-class taxonomy and the coarser four-class taxonomy that a repair stage consumes. Third, we establish by permutation analysis that the expert evidence streams, rather than the raw frame descriptors available to the same router, carry the routing decision. Fourth, we quantify the uncertainty of every reported gain with a case-level paired bootstrap and report per-recording variation, and we release a reference implementation that reproduces every number.

II. RELATED WORK

Classical restoration models clicks, pops, and scratches as localized impulsive disturbances. Statistical and autoregressive methods use prediction error or local signal models to find corrupted samples and then interpolate or suppress them [1], [2], and bidirectional processing improves pulse localization by combining forward and backward alarms [3]. These motivate our event-centered sparse branch, but their output is a repair mask rather than complementary regime evidence.

Supervised archival detection moves the decision earlier. Brandt et al. classify whether each 1 s frame contains impulsive disturbance, using prewhitening, so the resulting probabilities can gate a restoration algorithm and leave clean frames untouched [4]. Prior work separates sporadic clicks, crackle as a dense floor of small clicks, and low-frequency thumps, noting that dense bursts are perceived as crackle [5], [6]. These temporal structures motivate our event-level sparse branch and

frame-level burst and dense branches, extending binary gating to repair-oriented regimes. Perceptual-model detectors have likewise localized clicks in real analog excerpts [7].

Our feature choices also relate to transient analysis. Abrupt energy, spectral, and local statistical changes mark musical onsets as well as noise events [8], the central difficulty here, since high curvature, prediction error, or high-frequency energy is informative without proving a defect. Our front end therefore pairs LPC residual and curvature cues with shape, spectral, cepstral, density, and context statistics, and keeps candidate-level localization for the backend to interpret.

The repair target also determines what counts as useful detection. Audio inpainting reconstructs corrupted samples from surrounding structure [9], whereas crackle and overlapping damage need region-level denoising. Antiquing supplies a physical basis for synthesizing historical degradations [6], [10] and neural systems learn from them directly [11], but both target restoration quality rather than prerestoration routing. Our contribution is the intermediate representation, in which physically motivated experts produce separate evidence for later regime selection without forcing every artifact type through a single detector.

III. METHODOLOGY

The detector has two stages. A signal-informed front end estimates sparse, burst, and dense evidence, and a backend router then predicts

$$y_t \in \{\text{clean, sparse, burst, dense, mixed}\} \quad (2)$$

for each analysis frame t .

The two stages use different supervision units. Sparse damage is localized around event centers, whereas bursts and crackle occupy short or extended regions. Keeping these units separate preserves physical information, and the backend is the first stage that must emit one exclusive label per frame.

A. Benchmark Construction and Frame Targets

Each excerpt is corrupted by a seed-controlled recipe and stored with its clean and corrupted waveforms, sample-level masks, event table, and frame labels, giving both an inference input and an audit trail. Excerpt boundaries are fixed before corruption, and all feature and label arrays reset there.

The sparse branch is trained at candidate level, where a candidate is positive when its local maximum lies within 100 samples of a ground-truth event center. Burst and dense supervision are frame level and follow the supported-artifact logic for wide pops and short scratches and the dense mask-support criterion. For router training, candidate-level sparse evidence is projected onto frames. A frame is clean when no supported defect is active, mixed when dense and impulsive support coincide, and otherwise sparse, burst, or dense.

B. Signal-Informed Expert Front End

The waveform is converted to mono, median-centered, and normalized by a median absolute deviation estimate. An order-16 LPC residual and a second-difference curvature signal are

normalized the same way, and their positive components form the core defect score

$$s_{\text{core}}[n] = \sqrt{\max(s_{\text{LPC}}[n], 0) \max(s_{\Delta^2}[n], 0)}. \quad (3)$$

Local maxima define sparse candidates, described by residual and curvature magnitude, prominence, width, spacing, and density. Frame features use 1024-sample frames with a 512-sample hop and cover spectral flatness, high-frequency ratio, spectral spread, cepstral energy, crest factor, core-score statistics, peak count, and peak density.

Three binary MLP experts estimate complementary evidence, each with two ReLU hidden layers of 32 and 16 units, a sigmoid output, and standardization fixed from the training pool. The sparse expert operates on candidates and the burst and dense experts on frames. Their objectives are trained independently, so a strong response in one branch does not suppress another. Sparse candidate probabilities are aggregated per frame by the maximum, giving

$$\mathbf{p}_t = [P_s(t), P_b(t), P_d(t)]^T. \quad (4)$$

The outputs are evidence streams rather than mutually exclusive class probabilities, and mixed frames arise when dense support co-occurs with impulsive activity.

C. Hierarchical Temporal Router

For each frame the router receives a 101-dimensional vector

$$\mathbf{z}_t = [\mathbf{p}_t^T, \mathbf{c}_t^T, \mathbf{i}_t^T, \mathbf{u}_t^T, \mathbf{g}_t^T]^T, \quad (5)$$

where $\mathbf{c}_t$ holds the two largest expert probabilities, their margin, and the normalized posterior entropy, $\mathbf{i}_t$ holds pairwise expert differences and an agreement term, $\mathbf{u}_t$ holds local temporal statistics, and $\mathbf{g}_t$ holds raw frame descriptors together with candidate-density counts and their one-frame lag and lead. The temporal statistics are rolling means, standard deviations, extrema, first and second differences, and transition density of the expert trajectories over a centered window of ± 4 frames. The router is therefore non-causal by up to 46 ms, which suits an offline prerestoration pass. All windows are computed within an excerpt, so context cannot cross unrelated source segments.

Rather than one five-way classifier, the backend factors the decision into three softmax stages mirroring the physical taxonomy,

$$q_t(k) = \begin{cases} P(-\text{def}), & k = 0, \\ P(\text{def}) P(\tau_k | \text{def}) P(k | \tau_k), & k > 0, \end{cases} \quad (6)$$

where stage one separates clean from defective frames, stage two assigns a defect type $\tau \in \{\text{impulsive, dense, mixed}\}$, and stage three splits impulsive into sparse and burst, with $P(k | \tau_k) = 1$ for the dense and mixed types. The stages are ReLU MLPs with 28, 28, and 16 hidden units, each trained with class-balanced cross entropy on a capped per-class subsample, so the two majority regimes cannot dominate the macro-averaged objective. This lets the rare sparse/burst distinction be learned on impulsive frames alone rather than against 413,385 clean frames.

D. Mixed Calibration and Sequence Decoding

Mixed damage is treated as an overlap state rather than an independent morphology. The backend combines dense evidence with impulsive evidence

$$e_{\text{imp}}(t) = \max(P_s(t), P_b(t)), \quad (7)$$

and suppresses the mixed posterior by 0.02 on any frame where $P_d(t) < 0.75$ or $e_{\text{imp}}(t) < 0.65$. A second pass repeats this wherever the mixed posterior fails to exceed the dense posterior by -0.12 , removing mixed hypotheses that survive only because dense evidence is strong. Both thresholds are fixed on validation data. Given the resulting emissions $\tilde{q}_t(k)$, the final sequence minimizes

$$J(\mathbf{y}) = \sum_t -\log(\tilde{q}_t(y_t) + \epsilon) + \sum_{t>1} C(y_{t-1}, y_t) + \sum_t B(y_t), \quad (8)$$

where C penalizes changes and selected unstable transitions, and B is a per-frame class bias that discourages the mixed and isolated impulsive states.

The sequence follows from the standard Viterbi recursion. If $D_t(k)$ is the minimum cost of a partial sequence ending in state k , then

$$D_t(k) = -\log(\tilde{q}_t(k) + \epsilon) + B(k) + \min_j \{D_{t-1}(j) + C(j, k)\}. \quad (9)$$

Backpointers recover the minimum-cost sequence independently per excerpt, so the decoder suppresses isolated low-confidence flips while keeping a transition when accumulated emission evidence supports it.

All router, calibration, and decoder parameters are selected on a validation split of the training pool. The stay bonus is 1.50, switch penalty 0.25, mixed prior penalty 0.30, sparse/burst prior penalty 0.25, and the dense/mixed and clean/defect transition penalties both 0.10. Selection follows a tune-then-refit protocol. The penalties and mixed gate are chosen with a hierarchy fitted on the 418 fitting cases and scored on the 70 validation cases, then frozen, after which the hierarchy is refitted on all 488 development cases. The decoder is thus frozen before any held-out label is read, and test labels serve only the final metrics and the confusion matrix. The penalties therefore act on emissions from a model refitted on more data than selected them, which we revisit in Section V.

IV. EXPERIMENTS AND RESULTS

A. Benchmark and Protocol

Because real transfers cannot supply sample-accurate labels, we use a controlled synthetic benchmark built from clean music recordings, divided into deterministic non-overlapping 15 s excerpts at 44.1 kHz. Each realization injects one of five procedural recipes, namely narrow clicks, wide pops, short scratches, dense crackle, or mixed damage. The split is made at the level of the source recording. Eighteen development recordings yield 488 realizations and three held-out recordings yield 109, for 597 in total. The development pool is split by

TABLE I
FROZEN EXPERT RESULTS ON HELD-OUT TEST. SPARSE F1 IS CANDIDATE/EVENT LEVEL. BURST AND DENSE F1 ARE FRAME LEVEL. A ZERO ENTRY MEANS THE RECIPE CONTAINS NO POSITIVES OF THAT TYPE.

Defect recipe	Sparse F1	Burst F1	Dense F1
Narrow clicks	0.605	0.000	0.000
Wide pops	0.927	0.899	0.000
Short scratches	0.000	0.634	0.000
Dense crackle	0.000	0.000	0.986
Mixed damage	0.577	0.378	0.975

recording into 418 fitting and 70 validation cases. The held-out recordings are piano, orchestral, and vocal-jazz material sharing no source with the development pool. Their small number is the main limit on generalization, so every headline figure carries a case-level bootstrap interval.

All expert outputs are frozen for the backend comparison, and no held-out label is used for fitting, calibration, threshold selection, transition penalties, or decoder selection. We report two taxonomies. The five-class view separates sparse from burst and is the harder diagnostic target. The four-class view merges them into one impulsive class and matches what a repair stage consumes, since isolated clicks and short bursts are both handled by interpolation. Both are scored as pooled macro F1 over the aggregate confusion matrix, with weighted F1 also reported because clean and dense dominate the frames. Uncertainty is estimated by resampling whole excerpts with replacement over 4000 draws, which respects the correlation between frames inside an excerpt. System comparisons use the paired difference on the same resampled excerpts. No validation candidate satisfied the auxiliary clean, dense, and mixed floors at once. Once that set proved infeasible, the final rule was selected by the predefined validation-only penalized objective. This decision was made before any held-out test labels were read.

B. Frozen Expert Front End

Table I summarizes standalone expert behavior. Wide pops activate both the sparse and burst experts, short scratches mainly burst, and dense crackle is isolated cleanly by the dense expert. Mixed damage retains strong dense evidence while also activating both impulsive streams.

Overall sparse candidate F1 is 0.714, burst frame F1 is 0.511, and dense frame F1 is 0.961. A direct front-end evidence rule nonetheless reaches only 0.464 pooled macro F1 and produces 14,725 transitions, so the expert front end is physically meaningful but insufficient as a complete repair router.

C. Backend Results

Table II reports the main held-out results. Direct thresholding of expert evidence is far weaker than learned routing. A flat five-class router reaches 0.687 macro F1, and the hierarchy, temporal consistency, and Viterbi decoding raise this to 0.702, 0.713, and 0.730 (95% CI 0.694–0.761). The gain of the full backend over the flat router is +0.043 (95% CI +0.016 to

+0.070, $p = 0.001$). Decomposing it, the hierarchical factorization alone contributes +0.015, which the paired test cannot separate from zero (95% CI -0.008 to $+0.039$, $p = 0.20$), whereas calibration and sequence decoding contribute a further +0.028 (95% CI $+0.009$ to $+0.045$, $p = 0.003$). The practical gain therefore comes from temporal decoding rather than from the hierarchy on its own.

The same ordering holds under the four-class taxonomy, where the final system reaches 0.805 (95% CI 0.777–0.829) against 0.768 for the flat router, a gain of +0.037 ($p < 0.001$). That interval is the narrower of the two, because merging the two smallest-support classes leaves the repair-relevant claim better determined than the diagnostic one.

A natural alternative is to estimate the transition matrix directly. Decoding the same emissions with maximum-likelihood transitions and priors from the training labels gives 0.712 macro F1, below the 0.730 from validation-selected penalties (paired difference +0.018, 95% CI $+0.002$ to $+0.035$, $p = 0.025$). The maximum-likelihood decoder is the more conservative of the two, emitting 1,020 transitions against 1,476 and reaching a slightly better mixed F1, but it oversmooths sparse to 0.447. Empirical transition counts are dominated by the two majority regimes, so a small penalty set tuned against a macro-averaged objective transfers better than the likelihood fit.

A remaining question is whether the experts carry the routing information or whether the raw descriptors in $\mathbf{g}_t$ suffice alone. Permuting each feature group across frames and re-decoding over five repeats, the 50 expert-derived coordinates cost 0.519 macro F1 when destroyed, collapsing the system to 0.211, against 0.095 for the 39 raw descriptors and 0.074 for the 12 candidate counts. The expert streams are what the backend routes on, and the descriptors act as secondary context rather than a substitute, which is why the front end is worth training separately.

D. Per-Class Performance and Failure Analysis

Table III gives the final per-class metrics. Clean and dense reach 0.991 and 0.980 F1, burst 0.691 and sparse 0.565, and mixed remains the limiting class at 0.422.

Fig. 1 shows the remaining errors are structured. Sparse frames are taken for burst in 27.8% of cases and burst for sparse in 11.9%. Of mixed frames, 54.5% are predicted dense and 43.2% correctly, whereas only 0.6% of dense frames are predicted mixed. Mixed therefore carries dense evidence plus impulsive activity, unlike both a clean fifth morphology and an ordinary dense frame. It is also the least certain class, with 437 frames and a 95% interval of 0.352–0.494. Much of this comes from the label space rather than the features. Merging sparse and burst lifts impulsive F1 to 0.825, above either constituent class, so the evidence separating impulsive from dense activity is far stronger than that separating the two impulsive morphologies.

Temporal decoding cuts transitions from 14,725 under direct thresholding to 1,476, and from 2,119 relative to temporal consistency alone while macro F1 rises from 0.713 to 0.730, so

		Final Viterbi Decoder				
True regime	Clean	98.7% (91626)	0.1% (88)	0.0% (41)	1.1% (1030)	0.0% (3)
	Sparse	16.7% (74)	53.4% (236)	27.8% (123)	1.1% (5)	0.9% (4)
	Burst	9.6% (46)	11.9% (57)	77.0% (368)	0.6% (3)	0.8% (4)
	Dense	0.6% (298)	0.0% (12)	0.1% (46)	98.7% (45851)	0.6% (258)
	Mixed	0.0% (0)	0.2% (1)	2.1% (9)	54.5% (238)	43.2% (189)
		Clean	Sparse	Burst	Dense	Mixed
		Predicted regime				

Fig. 1. Row-normalized held-out TEST confusion matrix for the final Viterbi-decoded router. Clean and dense regimes are reliable. Mixed errors are dominated by confusion with dense, while sparse and burst remain partially ambiguous.

the decoder improves class interpretation as well as sequence stability.

E. Generalization Across Recordings

Because the held-out pool is small, Table IV breaks the final result down by source recording. Macro F1 ranges from 0.656 to 0.765 under the five-class taxonomy and from 0.740 to 0.819 under the four-class taxonomy. The weakest case is the vocal-jazz excerpt set, which is also the smallest at twelve realizations, and the strongest is the orchestral piano concerto. The spread is wide enough that a single additional recording could move the pooled figure by more than the gains reported above, which is why we treat the bootstrap intervals rather than the point estimates as the claim.

F. Reproducibility

The decoder, the calibration stages, and the evaluation are deterministic given the frozen models, so every number in this paper is reproducible from the released artifacts. A reference implementation independent of the original MATLAB code reproduces all reported metrics to within 5×10^{-16} , and it also regenerates the bootstrap intervals, the maximum-likelihood transition baseline, and the permutation analysis.

V. CONCLUSION

We presented a lightweight signal-informed system for prerestoration vinyl defect regime detection. Frozen sparse, burst, and dense expert evidence feeds a hierarchical router, a mixed-regime gate, and a validation-selected Viterbi decoder, reaching 0.730 five-class and 0.805 four-class macro F1, ahead of a flat router by 0.043 and 0.037. A paired bootstrap attributes most of that gain to calibration and sequence decoding rather than the hierarchy, and the selected penalties also beat

TABLE II
HELD-OUT TEST PERFORMANCE UNDER BOTH TAXONOMIES, CUMULATIVE BACKEND STAGES, ALL WITH FROZEN EXPERT EVIDENCE AND VALIDATION-ONLY SELECTION. FINAL ROW 95% BOOTSTRAP CI: 0.694–0.761 FIVE-CLASS, 0.777–0.829 FOUR-CLASS.

System	Five-class				4-cl.	
	Macro F1	Wtd. F1	Micro F1	Mixed F1	Macro F1	Trans.
Front-end evidence rule	0.464	0.926	0.895	0.051	0.597	14725
Flat five-class router	0.687	0.980	0.978	0.413	0.768	2439
+ hierarchy and mixed gate	0.702	0.982	0.981	0.335	0.775	2419
+ temporal consistency	0.713	0.983	0.982	0.367	0.785	2119
Viterbi, ML transitions	0.712	0.984	0.984	0.438	0.796	1020
Viterbi, selected penalties	0.730	0.983	0.983	0.422	0.805	1476

TABLE III
FINAL VITERBI PER-CLASS METRICS ON HELD-OUT TEST.

Regime	Precision	Recall	F1	Support
Clean	0.995	0.987	0.991	92788
Sparse	0.599	0.534	0.565	442
Burst	0.627	0.770	0.691	478
Dense	0.973	0.987	0.980	46465
Mixed	0.413	0.432	0.422	437

TABLE IV
FINAL SYSTEM BY HELD-OUT SOURCE RECORDING.

Source recording	Cases	5-class	4-class
Chopin, Ballade No. 4	50	0.712	0.803
Rachmaninoff, Piano Concerto No. 2	47	0.765	0.819
Fitzgerald and Armstrong, vocal jazz	12	0.656	0.740
Pooled	109	0.730	0.805

a maximum-likelihood transition matrix. Temporal structure rather than classifier capacity makes the evidence usable.

Four limits bound these claims. The taxonomy treats overlap as exclusive, tune-then-refit applies penalties to emissions from a refitted hierarchy, a held-out pool of three recordings is a narrow basis for generalization, and procedural degradation is not real vinyl wear. The first three need more data and a stricter protocol. The fourth is harder, because real transfers cannot supply the frame-level ground truth this evaluation depends on, so validation there needs region-level expert annotation and a downstream comparison in which routing selects the repair operator and is judged against a single-operator baseline. Transfer to an independent degradation model is a cheaper intermediate check.

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